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Problem

ONE PATH IS BRITTLE

- Greedy chain-of-thought decoding commits to a single locally likely rationale, which may contain an early mistake.
- Complex problems often admit multiple legitimate strategies that converge to the same answer.
- The challenge is to exploit this path diversity without training another verifier or collecting labels.

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Scope

- Evaluated with PaLM, GPT-3, LaMDA and UL2 across arithmetic and commonsense reasoning.
- No fine-tuning, auxiliary reward model or extra annotation is needed beyond the original CoT prompt.
- More samples generally improve accuracy with diminishing returns; diversity is essential, so beam-like paths help less.

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Method

SAMPLE, EXTRACT, AGGREGATE

- Use stochastic decoding to sample a diverse set of complete reasoning paths from the same CoT prompt.
- Extract the final answer from every path, discarding the wording and intermediate route for the decision.
- Return the most frequent answer; agreement implicitly marginalizes the latent reasoning paths.

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Evidence

LARGE CONSISTENT GAINS

- PaLM-540B gains +17.9 points on GSM8K, +11.0 on SVAMP and +12.2 on AQuA.
- Commonsense gains include +6.4 on StrategyQA and +3.9 on ARC-Challenge.
- Improvements hold across model families, prompt variations and sampling configurations, often setting new state of the art.

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Decision rule

CONSENSUS OVER RATIONALES

- $a_{\text{star}} = \operatorname{argmax}_a \sum_i 1[\text{answer}(r_i) = a]$
- Paths that make different mistakes tend to disperse their votes, while correct reasoning often converges.
- Unweighted majority is simple and strong; probability-weighted variants are also explored.

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Limits & meaning

ACCURACY FOR MORE COMPUTE

- Sampling many long rationales multiplies latency and token cost compared with greedy CoT.
- Correlated model errors can create a confidently wrong majority; answer extraction can also fail on open-ended outputs.
- The method established test-time reasoning compute as a powerful axis: search over thoughts can improve a frozen model.

SELF-CONSISTENCY

Self-Consistency Improves Chain-of-Thought Reasoning

Sample Diverse Paths, Marginalize the Answer | ICLR 2023

Do not trust one greedy rationale: sample many diverse chains of thought and choose the answer reached most consistently, marginalizing over different valid reasoning

sampling

answer voting

paths.

test-time compute